Pavni Arora

B.Tech Information Technology — VIT Vellore
pavniarora16@gmail.com +91 8604459554 LinkedIn GitHub

Summary

Undergraduate B.Tech IT student at VIT Vellore with a CGPA of **9.54**, strong foundations in **Python, Java, C and C++**, and hands-on experience in **data structures, algorithms, computer vision, and research-oriented projects**. Actively building algorithmic solutions and scalable software systems.

Education

Vellore Institute of Technology (VIT), Vellore 2024 – Present
B.Tech in Information Technology
CGPA: 9.54 *8th Rank in Branch*

Technical Skills

Programming: Python, Java, C, C++, Web Programming (HTML, CSS, JS, JQuery, JSON, AJAX, ReactJS, AngularJS, ExpressJS, NodeJS, MongoDB)
CS Fundamentals: Data Structures & Algorithms (Arrays, Strings, Linked Lists, Stacks, Queues, Trees – ongoing)
AI / Computer Vision: OpenCV, CNNs (conceptual), Transformer architectures (theoretical)
Tools: Git, GitHub, VS Code

Projects

Personal Portfolio Website *HTML, CSS, JS, ReactJS, ExpressJS, AngularJS, NodeJS*

- Designed and developed a responsive portfolio website to showcase projects and achievements.
- Implemented clean UI layout, navigation, and reusable components.

Deepfake Detection System (SIH Project) *Python, Computer Vision, Deep Learning*

- Developed a deepfake detection system to identify AI-generated images and videos.
- Trained a machine learning model on image datasets to classify real vs manipulated media.
- Applied computer vision techniques for feature extraction and model evaluation.

Huffman Coding – File Compression Tool *Python*

- Implemented Huffman Encoding algorithm to achieve efficient lossless file compression using variable-length codes.
- Built frequency maps, binary trees, and encoding/decoding logic using core data structures.
- Demonstrated strong application of greedy algorithms and tree-based data structures.

Gesture-Controlled Snake Game *Python, OpenCV*

- Developed a real-time gesture-controlled Snake game using computer vision techniques.
- Implemented hand tracking via webcam for smooth and responsive gameplay.

Hangman Game *Python*

- Designed structured control flow and efficient string handling for game logic.
- Implemented win/loss state management and robust input validation.

Priority-Based Emergency Interrupt System (CAO Course Project) *Arduino, Embedded Systems, Python (PySerial)*

- Designed and implemented a hardware-software system to simulate priority-based interrupt handling using Arduino.
- Modeled real-time emergency scenarios with multi-level priorities and distinct system responses using LEDs and buzzer alerts.
- Demonstrated key Computer Architecture concepts including interrupt handling, polling, and I/O operations.

E-Shopping Cart (Web Technologies Course Project) *HTML, CSS, JavaScript, JQuery, JSON, AJAX, AngularJS, ExpressJS, NodeJS*

- Developed a responsive e-commerce web application with an interactive product catalog and shopping cart functionality.
- Implemented dynamic cart management, including item addition, removal, quantity updates, and real-time price calculation.
- Enhanced user experience through intuitive navigation, responsive design, and seamless client-side interactions.

Research & Technical Work

Training an **object detection model** for ISDC 2026 (Software Domain), involving dataset preparation and model evaluation.

Authoring a **review paper** comparing CNN and Transformer-based architectures for computer vision tasks (under review).

Developing a **disease-specific protein–protein interaction (PPI) network** by integrating STRING and PSICQUIC datasets and applying network analysis and clustering techniques to identify biologically significant gene modules (ongoing).

Achievements

Certificate of Merit: Ranked among the **Top 10 students** twice in the IT branch.

Code Golf Contest – **3rd Place** (IEEE WIE): Solved algorithmic problems emphasizing logic optimization and minimal code length.

Business Blaze Ideathon – **1st Place**: Proposed **LeadHers**, a platform addressing funding and mentorship gaps.

Leadership & Activities

Vice Chairperson – **IEEE Women in Engineering (WIE), VIT**: Leading chapter initiatives, coordinating technical events and hackathons, and promoting women's participation in technology and engineering.

Event Coordinator – **Switchuation 2.0** (IEEE WIE Hackathon).

Workshop Facilitator – **IEEE Women in Engineering**: Conducted HTML/CSS workshop with a live mini-project.